

EngTrace Taxonomy

1 Chemical Engineering

**Transport Phenomena**

Shell Momentum Balances

Viscosity & Momentum Transport

**Reaction Kinetics**

Mole Balances

Stoichiometry

Conversion & Reactor Sizing

**Thermodynamics**

Heat Effects

Volumetric Properties of Pure Fluids

2 Electrical Engineering

**Signals & Systems**

Discrete Time Signals

Continuous Time Signals

**Electromagnetics & Waves**

Electrostatics

Magnetostatics

Waves & Phasors

**Digital Communications**

Deterministic & Random Signal Analysis

Digital Modulation Schemes

3 Civil Engineering

**Structural Analysis**

Determinate Structures

Influence Lines

Deflections

Indeterminate Analysis

**Geotechnical Engineering**

Phase & Index Properties

Permeability, Seepage & Effective Stress

Stress Distribution & Consolidation

Strength & Stability

**Water Resources**

Uniform Flow

Energy & Rapidly Varied Flow

Hydrology

4 Industrial Engineering

**Production & Inventory**

Deterministic Lot Sizing

Stochastic Inventory

Production Planning

**Quality & Reliability Control**

Variables Control Charts

Attributes Control Charts

Process Capability

Acceptance Sampling

**Stochastic Operations**

Queueing Systems

Markov Chains

Poisson Processes

System Reliability

5 Mechanical Engineering

**Fluid Mechanics**

Fluid Statics

Fluid Kinematics

**Mechanics of Materials**

Torsion

Stress & Strain

**Vibrations & Acoustics**

Harmonically Excited Vibrations

Single Degree of Freedom Systems

